Reinforcement Learning

HSE, winter - spring 2026

Lecture 11 [Part II]: RL in Finance

Viacheslav Buchkov

RL in Finance

What Makes Finance Different?

Learning to act by sequentially *interacting* with an uncertain environment

As we have noticed during the course, the prevailing idea of RL was to solve a very general task with little domain knowledge, but many interactions.

However, financial applications usually:

- Have very **low signal-to-noise ratio** → POMDP;
- **Sparse and delayed rewards** → daily predictability ≠ monthly predictability;
- Observed d^{μ} is **non-stationary over time** → not a lot of data to train on;
- Simulation is **fully offline** → in case our actions affect d^{π}, we can deviate far from our simulator, where we do not have reliable transitions.

Dense or Sparse?

q_t

- Many RL approaches to finance argue that we have a **dense reward feedback**, i.e., we can observe a return every millisecond;
- However, we know that in finance **predictability is not homogeneous across different time periods**:
  - Orderbook features are good intraday predictors, but bear little information for monthly predictions;
  - Macro (e.g., unemployment) is a slow signal, so daily predictability usually doesn't work;
  - Moreover, a weak daily signal has usually lower predictability than the transaction costs you pay.
- Therefore, either we resample a **dense feedback to the desired time period** or we **treat a reward as a sparse one**

1m

Portfolio Management

Lecture 8

$$\begin{array}{cc} & K \\ d & \left\{\begin{array}{ccc} w_{1,1} & \cdots, & w_{1,K} \\ & \vdots & \end{array}\right. \end{array}$$

Dynamic Portfolio Management Task

Learning to act by sequentially *interacting* with an uncertain environment

Given an initial wealth M_0, investors want to choose the consumption c_t and portfolio weights adjustment Δw_t to maximize their long-term wealth

$$V_t(M_t, w_t) = \max_{c_t, \boxed{\Delta w_t}} [\boxed{u(c_t M_t)} + \gamma \mathbb{E} V_{t+1}(M_{t+1}, w_{t+1}])]$$

By definition the problem looks exactly like an RL problem:

- $a_t = (c_t, \Delta w_t)$, $s_t = (M_t, w_t)$

M_t

- State-transitions:
  - $M_{t+1} = b_t M_t R_f + ((w_t + \Delta w_t) M_t)^T \mathbf{R}_t$

  w_{t+1}

  - $$w_{t+1} = \frac{(w_t M_t + \Delta w_t M_t) \odot \mathbf{R}_t}{M_{t+1}}$$

But where is the problem?

$O_t \to \pi(O_t)$

Learning to act by $\boxed{\text{sequentially}}$ *interacting* with an uncertain environment

Given an initial wealth M_0, investors want to choose the consumption c_t and portfolio weights adjustment Δw_t to maximize their long-term wealth

$$V_t(M_t, w_t) = \max_{c_t, \Delta w_t}[u(c_t M_t) + \gamma \mathbb{E} V_{t+1}(M_{t+1}, w_{t+1}])]$$

By definition the problem looks exactly like an RL problem:

- $a_t = (c_t, \underline{\Delta w_t}), s_t = (M_t, w_t)$

$R_{t+1} = \frac{P_{t+1}}{P_t} - 1$

GBM

- State-transitions:
  - $M_{t+1} = b_t M_t R_f + ((w_t + \Delta w_t) M_t)^T \boxed{\mathbf{R}_t}_{+1}$
  - $w_{t+1} = \frac{(w_t M_t + \Delta w_t M_t) \odot \boxed{\mathbf{R}_t}_{+1}}{M_{t+1}}$

$\Longrightarrow$ use a simulation?

see theoretical approach [here]

$\min_{w} \|w^{opt} - w\|_2^2 - \lambda \|w - w_{t-1}\|_2^2$

Contextual Bandits Relaxation

- The sequential dependence comes **exclusively from the transaction costs:**

✓ • $b_t M_t = (1 - \mathbf{1}^T w_t) M_t - \mathbf{1}^T \Delta w_t M_t - \mathbf{1}^T (\tau \boxed{|\Delta w_t|} M_t) - c_t M_t$

- **Solution** — use **augmented utility** $\tilde{u}(c_t M_t) = \underline{u(c_t M_t)} - \underline{\lambda^{CV} \tau |\Delta w_t|}$

- Then we can avoid state-transition dependence and treat the problem as a contextual bandits task

Markowitz

- The usual choice is quadratic utility $u^{[a,b]}(c_t M_t) = a(c_t M_t) - b(c_t M_t)^2$

$$\pi^*(s) \in \underset{a}{\text{argmax}}[r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' | s, \cancel{a}) V^*(s')] = \underset{a}{\text{argmax}}\, r(s,a)$$

$$\underline{\hat{\pi}(s) \sim \hat{r}(s,a)}$$

$$w^T\mu_{t+1} - \frac{\gamma}{2} w^T \Sigma_{t+1} w$$

Contextual Bandits Portfolios

Intermezzo: Supervised Learning Relaxation

- Actually we can show that the portfolio management problem can be approximated as a supervised learning task!

$\| w_t \|_1 \leq R$

- The PnL of portfolio is $\boxed{\langle w_t, R_{t+1} \rangle}$ we have a maximum GMV $\| w_t \|_2 \leq C$

- Applying Cauchy-Schwarz we can see

$$|\langle w_t, R_{t+1} \rangle| \leq \| w_t \|_2 \| R_t \|_2 \leq C \| R_t \|_2$$

- Our collected return is proportional to the return in that asset $w_t \propto R_t$

- Thus, by C-S the maximum is attained, whenever the rhs is equal to the lhs

- If we have a **prediction of $\mathbf{sign}(\mathbf{R_t})$**, we should allocate the whole capital in the direction of return $w_t = C \dfrac{R_t}{\| R_t \|_2}$ (a vector on the unit spehere)

Intermezzo: Supervised Learning Relaxation

Why then would we need the contextual bandits formulation?

Reward Function (not all utilities are simply PnL maximization)

Uncertainty in our decision making!

Uncertainty-Aware Portfolios

ML for Asset Pricing: The Tale of Uncertainty

Can we improve our decision-making under uncertainty?

- Usually, the ML for Asset Pricing literature focuses on **supervised** prediction modelling with **heuristic** portfolio construction (for example, see, Gu, Kelly, and Xiu 2020, Shen and Xiu 2025)

- The literature is dominated by the two-stage factor construction: sorting on signal and taking the positions equally-weighted

POMDP

- While parameter-based portfolio construction methods are well-known, we usually **do not account for the model uncertainty**

- We suggest to ask: is there a way to **account for the fact that not all the stocks are equally-well predictable**?

Setup

- We focus on the **monthly** rebalanced portfolios from 2010/01/01 to 2024/12/31
- We work with the CRSP Dataset, filtered by QSEC US universe
- Today we want to focus on the approach, rather than feature engineering, so we use pre-built set of features
- Our features are month-end stock characteristics from Jensen, Kelly, Pedersen (jkpfactors.com) through CRSP: $X_{t,k} \in \mathbb{R}^{403}$
- We transform the features into cross-sectional z-scores: $Z_{t,k} = \dfrac{X_{t,k} - \bar{X}_t}{\sigma_t}$
- The targets are clipped excess returns: $Y_{t,k} = \text{clip}(R^e_{t,k}, R^e_{t,k} + 2\tilde{\sigma}_t, R^e_{t,k} - 2\tilde{\sigma}_t)$

Setup

Sorted Portfolios:

- We end up with a high-dimensional problem P = 403 versus N = 10×12×1,700
- $Y_{t,k} = f(Z_{t,k}) + \epsilon_{t,k}$
- Following the discussion in Shen and Xiu 2025, we treat the problem as **dense**, not sparse, meaning that **most of the stock characteristics retain some signal**
- As each characteristic is a sorting variable in the sense of classical factor construction, this is a reasonable assumption
- Therefore, we apply Ridge in the supervised learning task:
- $Y_{t,k} = Z_{t,k}\theta + \epsilon_{t,k},\ ||\theta||_2 \leq \tau$

Baseline Approach

Sorted Portfolios:

- The usual approach in Financial ML literature (see overview in Kelly and Xiu 2023) would be to apply ranking on the model predictions
- Define cross-sectionally ranked prediction as $\mathscr{R}(\hat{R}_{t,k}) \in [0\,\%\,,\ 100\%]$
- Employ a simple portfolio construction rule:

$$W_{t,k} = \begin{cases} 1 & \mathscr{R}(\hat{R}_{t,k}) \geq q\% \\ -1 & \mathscr{R}(\hat{R}_{t,k}) \leq q\% \\ 0 & \text{otherwise} \end{cases}$$

$1/\Sigma 1$

$-1/\Sigma \mathbb{1}_{[=-1]}$

- For hedging we use daily rebalanced ES1 Futures

Contextual Bandits Portfolios

Let's formulate our task as a Contextual Bandits Task:

- What if in our panel the stocks are not equally-well predictable by the *fitted* ML model? (**Inductive Bias 1**)
- We would like to **account for this fact in our decision-making directly**
- To simplify our learning task, we stick to equally-weighted portfolios in each leg (**Inductive Bias 2**)

Contextual Bandits Portfolios

- State $s_{t,k}$ is the unobservable true "company business" state
- We work under a low signal-to-noise ratio (**Inductive Bias 3**), so we observe only a context:
  - True Features: $\tilde{X}_{t,k} = X_{t,k} \cup U_{t,k}$
  - We observe only a context: $X_{t,k}$

$w \in \mathbb{R}$

$w < 0 \Rightarrow$ short

- For each stock at each month we pick an action, which weight we assign:

$$a_{t,k} \in \mathscr{A} = \{w = 1, w = -1, w = 0\}$$

- We treat our "reward" as dependent on unknown function $g : \mathbb{R}^{403} \rightarrow \mathbb{R}$

$$r(\tilde{X}_{t,k}, a_{t,k}) = a_{t,k} \cdot g(\tilde{X}_{t,k}) + \epsilon_{t,k}$$

Contextual Bandits Portfolios

Learning to act by sequentially *interacting* with an $\boxed{\text{uncertain environment}}$

- For each stock at each month we pick an action, which weight we assign:

$$a_{t,k} \in \mathscr{A} = \{w = 1, w = -1, w = 0\}$$

- We treat our "reward" as dependent on unknown function $g : \mathbb{R}^{403} \rightarrow \mathbb{R}$

$$r(\tilde{X}_{t,k}, a_{t,k}) = a_{t,k} \cdot g(\tilde{X}_{t,k}) + \epsilon_{t,k}$$

- The core idea is that we have an "abstention" action $w = 0$

- Therefore, **whenever our model has high uncertainty** about the prediction, **we should "abstain"** from acting, i.e., just not include the stock in the portfolio

Remember Disjoint LinUCB

- We can make use of Bayesian Statistics and obtain a **probabilistic regressor**
- Treat the Ridge weights as random variables: $\theta \sim \mathcal{N}(0, \tau^2 \mathbf{I})$
- With noise of the residual σ^2 we have penalty parameter $\lambda = \dfrac{\sigma^2}{\tau^2}$ (where $\sigma^2_{CV} \implies \lambda_{CV}$)
- Posterior predictive:
- $\mu = (X^T X + \lambda \mathbf{I})^{-1} X^T y$
- $\Sigma = \sigma^2 (X^T X + \lambda \mathbf{I})^{-1}$
- $\hat{R}_{t,k} \sim \mathcal{N}({x^*}^T \mu, {x^*}^T \Sigma x^*)$

Decision Rule

As we now obtained the whole **distribution** approximation instead of the (MAP) point prediction, we can construct a **Bayes-optimal decision rule** as:

Algorithm 1 Uncertainty-Aware Portfolio

$LCB_{t,k} = \mu(X_{t,k}) - \beta\sigma(X_{t,k})$

$UCB_{t,k} = \mu(X_{t,k}) + \beta\sigma(X_{t,k})$

procedure DECISION

    $LCB_{t,k} > 0 \rightarrow$ **confident enough to go long** $\Rightarrow w = 1$

    $UCB_{t,k} < 0 \rightarrow$ **confident enough to go ~~long~~** short $\Rightarrow w = -1$

    otherwise **abstain** $w = 0$

end procedure

Decision Rule

How do we pick β?

- $LCB_{t,k} = \mu(X_{t,k}) - \beta_t \sigma(X_{t,k})$
- $UCB_{t,k} = \mu(X_{t,k}) + \beta_t \sigma(X_{t,k})$

- Match the heuristic selection idea from the literature by making sure to select $\geq 2q\,\%$ of stocks in the universe
- Calibrate β_t to have abstention $\{LCB_{t,k} < 0\} \cap \{UCB_{t,k} > 0\}$ of $\approx 50\%$ with Binary Search
- Thus, we end up with the selected stocks of the same order as the comparable methods in the literature

Simulation Study

- Use a noisy version of leaked $R_{t,k} + \epsilon_{t,k}^{(i)}$ as a signal for predicting $R_{t,k}$ with $\epsilon_{t,k}^{(i)} \sim \mathcal{N}(0,\sigma^2)$

  $R_{t+1,k}$

- Create p such noisy versions of returns and $1 - p$ pure noise $\epsilon_{t,k}^{(i)}$

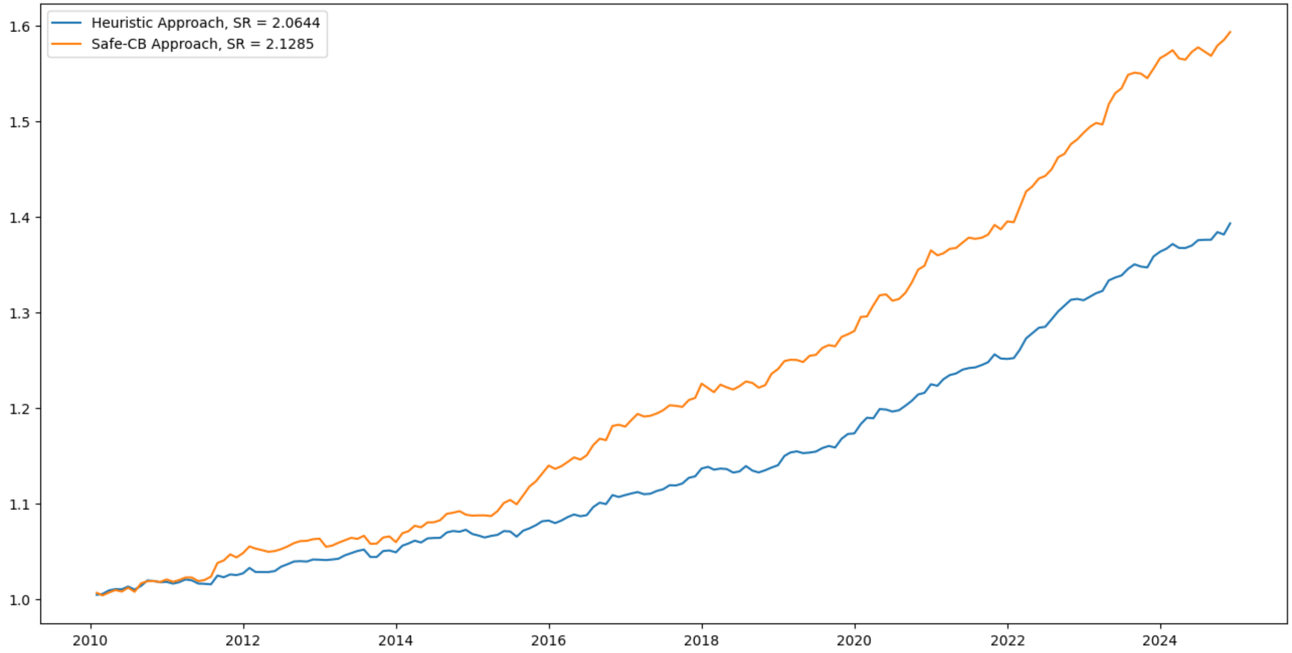
- Concatenate all these features to end up with $X_{t,k}^{sim} \in \mathbb{R}^P$

- We fix total number of features $P = 500$ to match the factor construction problem

  493

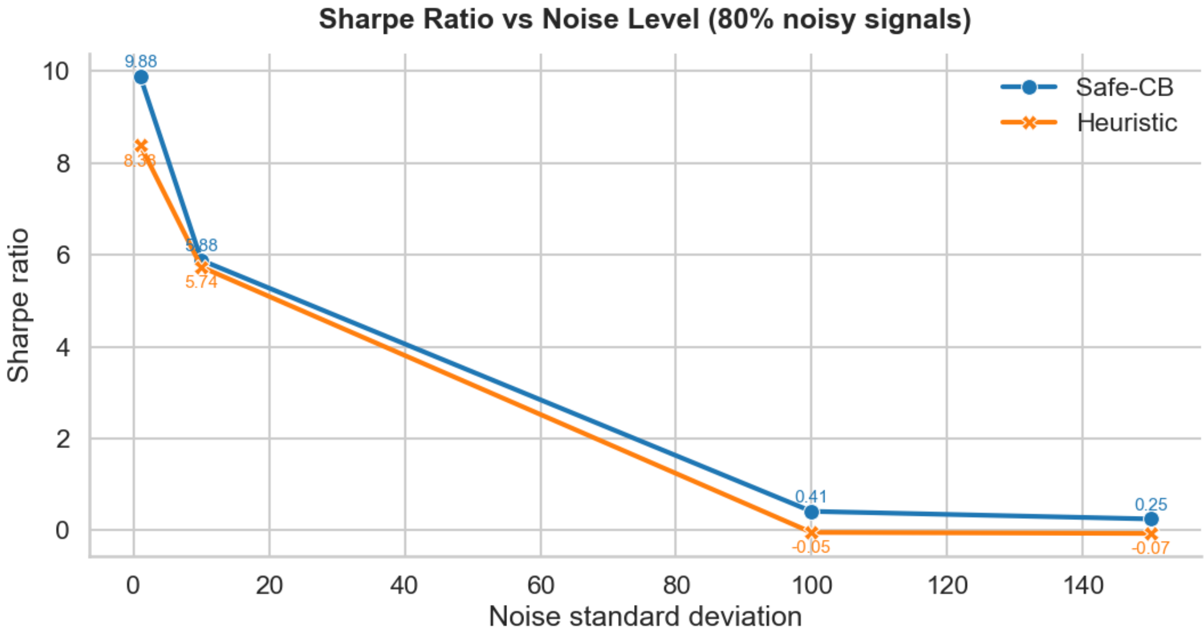
Simulation Study

Vol-Scaled, 100.0 Noise, 0% Noisy Signals, 2010/01-2024/12



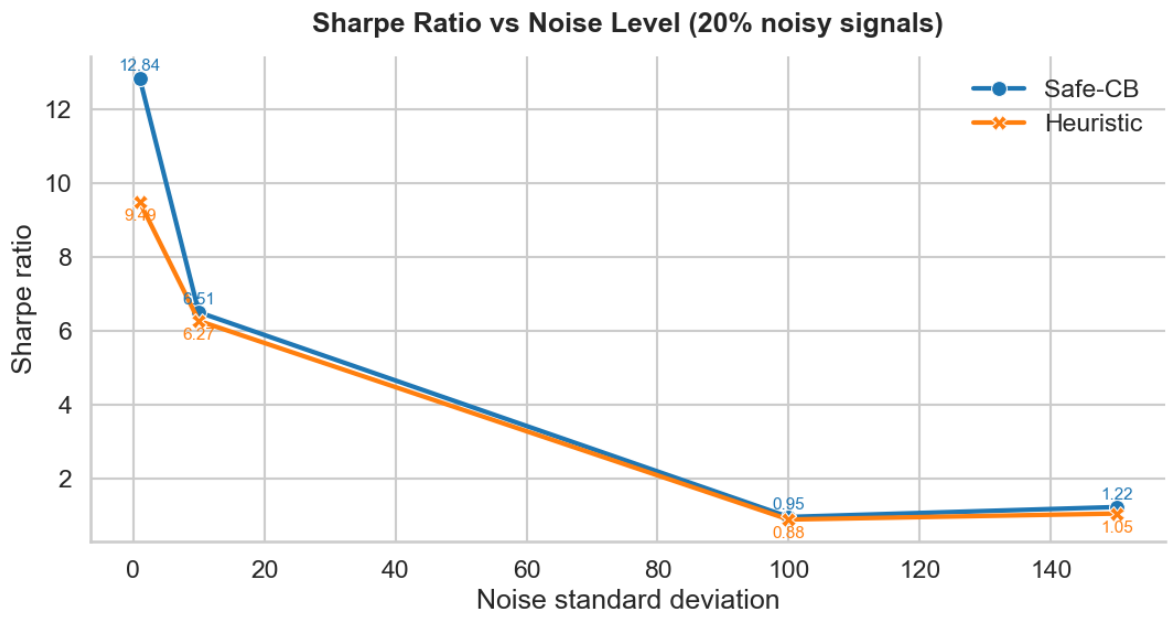
Simulation Study

80% Noisy Signals, 2010/01/01-2024/12/31



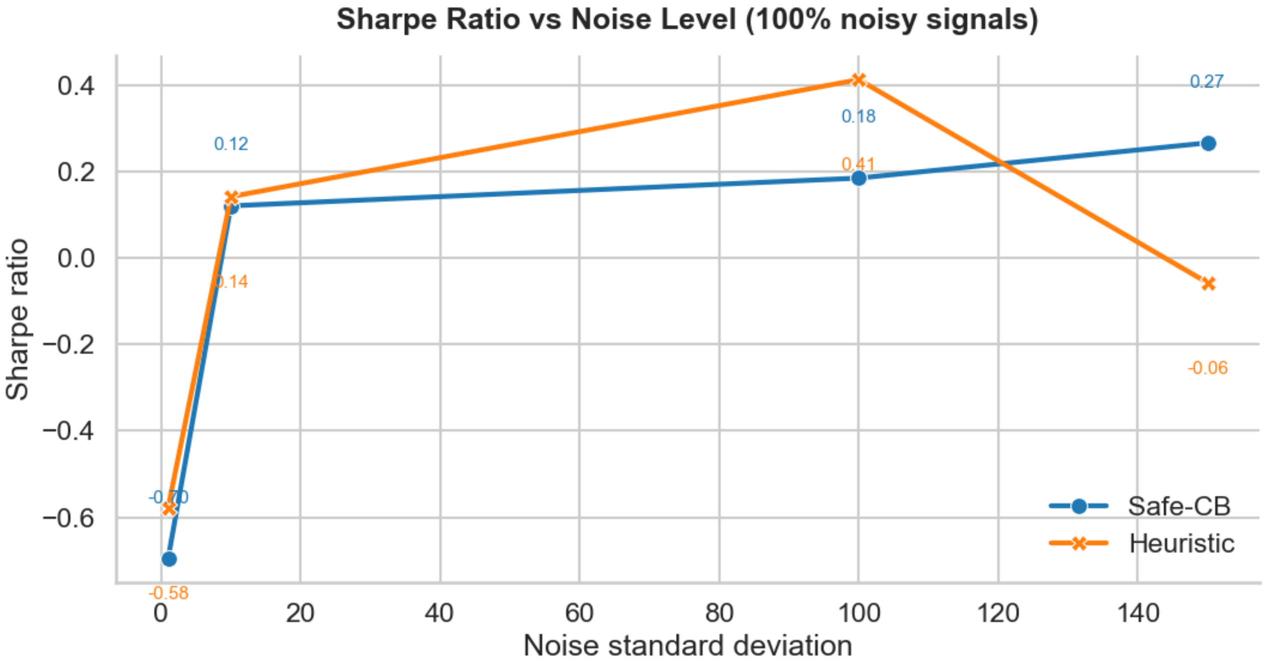
Simulation Study

20% Noisy Signals, 2010/01/01-2024/12/31

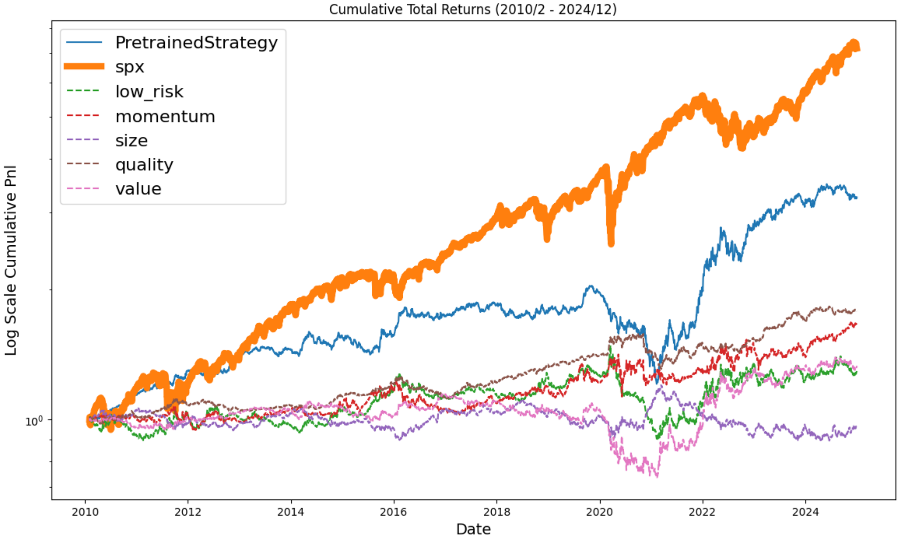


Simulation Study

100% Noisy Signals, 2010/01/01-2024/12/31

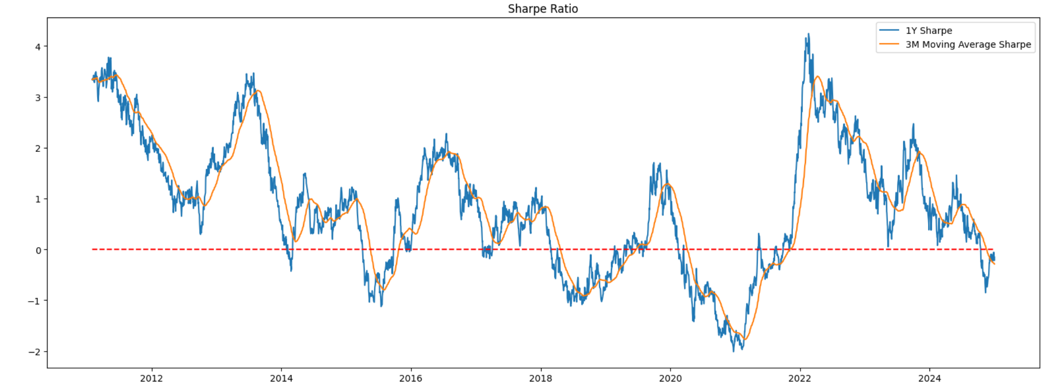


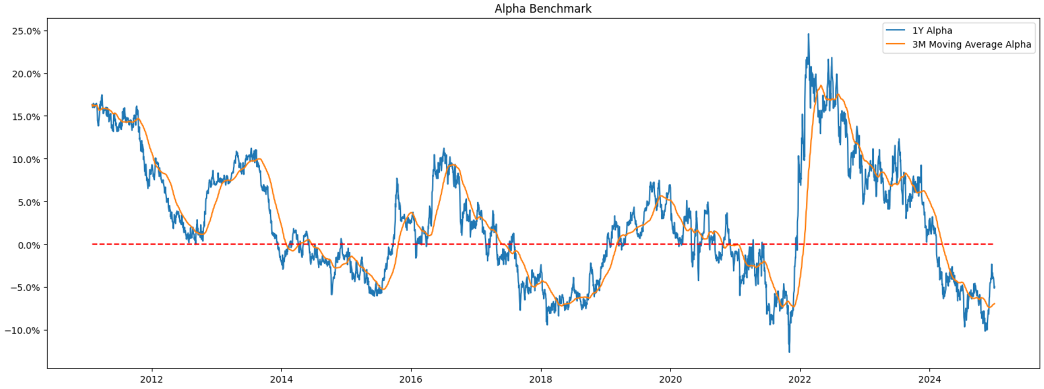
Strategy Performance



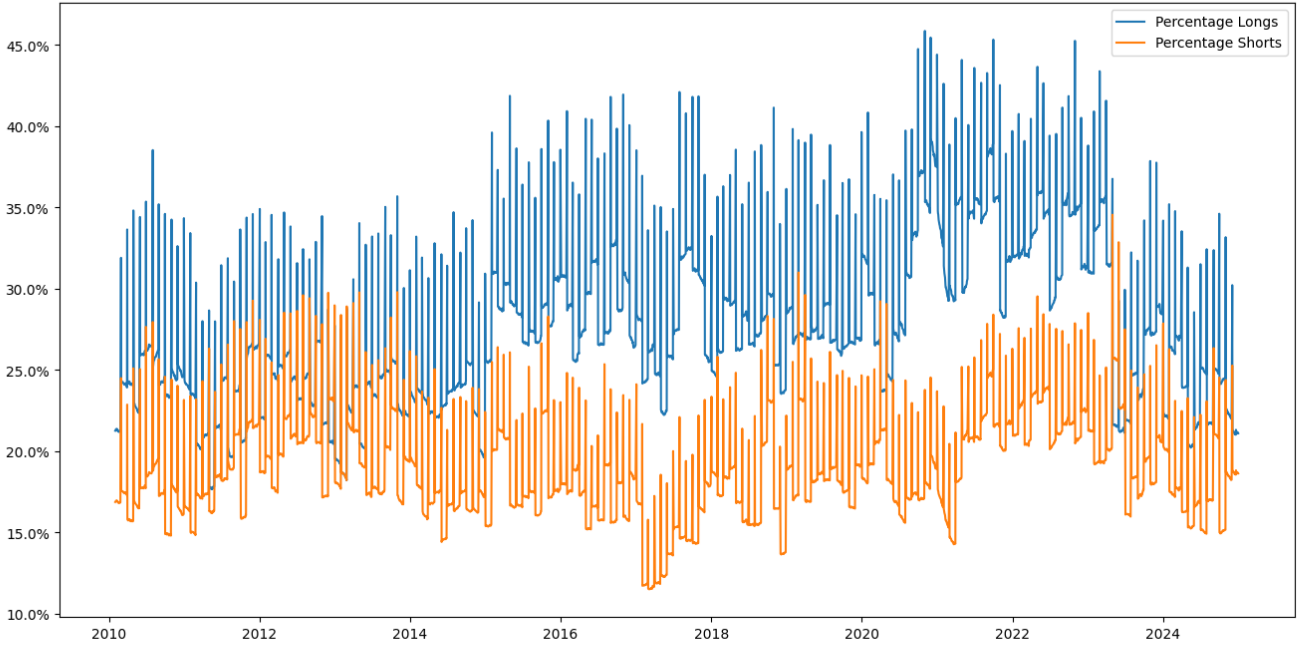
Strategy	SR	Ann. Ret	MDD	Ann. Alpha	IR	Acc
Ridge	0.4347	4.27%	**-25.00%**	0.60% (0.54)	0.1458	51.68%
CB	**0.6107**	**8.25%**	-41.32%	**2.50% (0.14)**	**0.3742**	**52.72%**

Rolling Metrics





Legs Imbalance



Summary

- By leveraging the uncertainty-aware decision-making we have been able to **improve the Sharpe Ratio from 0.43 to 0.61**
- As feature engineering was not the goal at the current stage, results are driven by overall complexity of the panel predictions
- According to simulation studies, more skillful models can expect to improve the base strategy more significantly
- Random Forest? **Natural uncertainty approximation through ensembling**
  → Similar (Bernoulli) derivations for classification task
- Neural Networks? Treat $\theta_i \sim \mathcal{N}(0,\tau^2)$
  → Promising results, but tough to calibrate, work in progress
- The **uncertainty-aware decision rule is virtually model-free**, so it can be applied to any ML Factor Strategy

Optimal Strategy Allocation

Lecture 5

Futures vs ETF Stat Arb

The Futures (excluding funding gains) and ETF Prices are converging / diverging, based on funding cost changes and the market participants actions.

Futures vs ETF Stat Arb

- One **cannot short the ETF** (in white) to close up the arbitrage

- However, imagine that a portfolio manager has some capital C to allocate into the market exposure

- She is indifferent between Futures and ETF, **as long as she doesn't underperform the ETF**

- Use RL for capital allocation task!

Futures vs ETF Stat Arb

- We have some features X_t that describe market conditions
- Our task is to choose on each day a capital portion $a_t \in \mathbb{R}_+$ to allocate into the Futures, until the whole budget is exhausted $\sum_{t=0}^{T} a_t \leq C$
- Introduce a state $s_t = \{X_t, C - \sum_{t' \leq t} a_{t'}, T - t\}$
- Treat the reward as a simplified outperformance $r_t = a_t \cdot p_t \cdot (T - t)$, as p_t is known at the time we decide how much a_t to allocate
- However, we do not know all future p_t (otherwise, a so-called "bang-bang" task)

Futures vs ETF Stat Arb

- We trade-off the current (positive or negative) outperformance versus future higher $p_{t'}$ that we could potentially collect

Theorem (Bellman):

Policy is optimal $\iff$ it is greedy w.r.t. its induced Value function

$$\pi^*(s) \in \underset{a}{\text{argmax}}[\underline{r(s,a)} + \gamma \underline{\sum_{s' \in \mathcal{S}} P(s' \mid s, a) V^*(s')}] = \underset{a}{\text{argmax}}\, Q^*(s,a)$$

- Interestingly, here you do not need to learn well the p_t — if you just known that there may be spikes up in future, it is **enough to improve the baseline of investing the whole capital**, whenever $p_t > 0$

Futures vs ETF Stat Arb

Ways to approach this task:

- Bayesian MPC (learn path-dependent p_t with uncertainty)
- SAC-Lagrangian or PPO-Lagrangian or TRPO/CPO

Challenges:

- Lack of full episodes (quarterly futures => only 4 futures * 5 years = 20 episodes) however, could rewrite to overlapping futures daily allocations (252 days * 5 years data)
- Feature engineering requires some expertise, however, a simple model would still outperform a simple heuristic rules with high probability (Bayes-optimal rule under uncertainty)

Optimal Execution

Lecture 5, Lecture 7

Optimal Execution

Selling Task: $\underset{q_1,\ldots q_T}{\operatorname{argmax}} \sum_{t=1}^{T} (q_t \cdot p_t)$, s.t. $q_t \in \mathbb{R}_+$, $\sum_{t=1}^{T} q_t = Q$

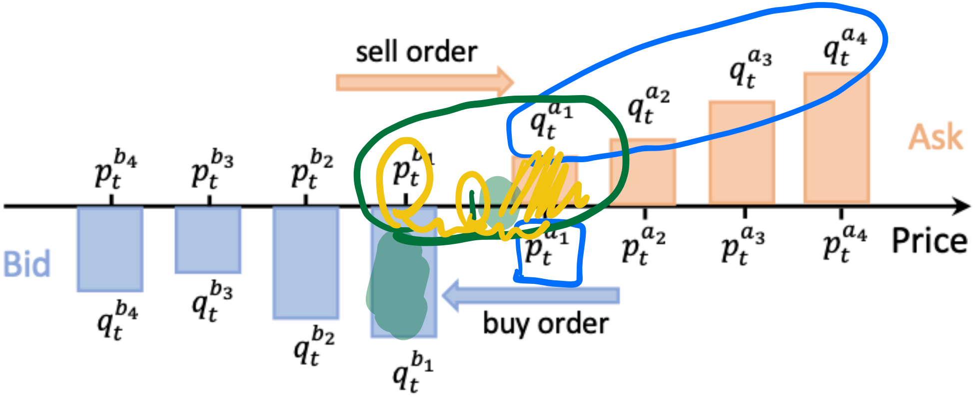


Fig. 3. A Snapshot of Limit Order Book.

Source

Another Example

Optimal Execution

Ways to approach this task:

- Usually, a finite action space task (at which levels which quantities to allocate)
- Papers include Q-Learning, DQN, DDQN, PPO Source

Advantages of RL:

- Sequential decision-making with relatively simple baselines to improve
- Multiple datapoints

Challenges:

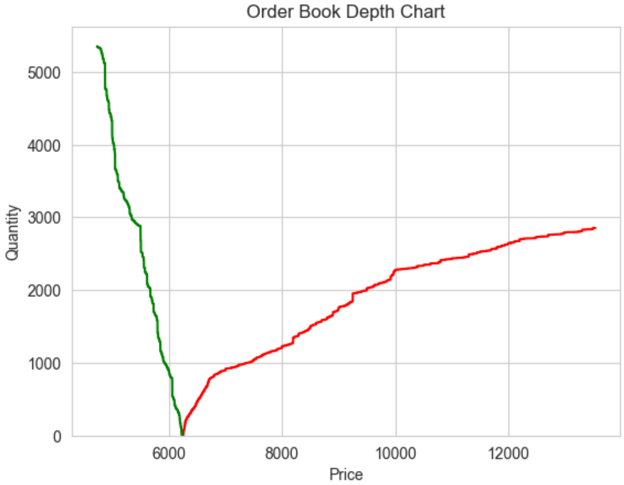
- The observed behaviour of other agents is not equivalent to the one we would've observed with our orders in the orderbook — attempts to solve with simulation [Byrd, et al. 2019], [Vyetrenko et al. 2019] or apply IQL / CQL

Market-Making

Lecture 7, Lecture 8, Lecture 9

Market-Making

Market-Making Task: $\operatorname{argmax}_{q_1,\ldots q_T} \sum_{t=1}^{T} (q_t \cdot p_t)$, s.t. $q_t \in [-C, C]$



Order Book	SHARES	PRICE
ASKS	22	69900
	17	69800
	140	69700
	24	69600
	6	69500
BIDS	42	69300
	42	69200
	41	69100
	32	69000
	21	68900

Fig. 3. Limit Order Book

Source

Market-Making

Ways to approach this task:

- Usually, a finite action space task (at which levels which quantities to allocate)
- Papers include Q-Learning, SARSA, PPO, Model-Based Source

Advantages of RL:

- Sequential decision-making, multiple datapoints and high quality features

Challenges:

- Same as OE: the observed behaviour of other agents is not equivalent to the one we would've observed with our orders in the orderbook**, even more critical here**   Multi-Agent RL $\rightarrow$ Markov Game
- Sparse Rewards: we collect a spread-paid from the market participants, but we take upon the **inventory risk**, which will be **realized in the future**

Market-Making: Example of Reward Design

How should we tackle this challenge? Let's design a potential function

- We have two sources of risk: **spread-crossing** (need to liquidate our current inventory) and **MtM** (the mid-price changes and your inventory is reevaluated)
- For example, we can introduce $\Phi(s_t) = -\beta h_t |I_t| - \alpha\sigma_t^2(T-t)I_t^2$ (based on the quadratic market impact assumption)
- Introduce reward shaping (under $\gamma = 1$): $\tilde{r} = r_t^{base} + \Phi(s_{t+1}) - \Phi(s_t)$
- Main challenge is **creating a reliable potential function / model-based setup**

Market-Making: Contextual Quoting

Another dimension of RL for Market-Making

- Usually, market-makers quote different instruments on OTC
- In these trades you know your counterparty outright, instead of treating the whole market as one “representative agent”
- You would like to avoid “adverse selection” — i.e., quoting a tight bid-ask spread to the hedge fund that **possibly has a good prediction on price change**
- **Solution**: Model this task as a contextual bandits / <u>counterparty-dependent RL</u> (adding the features of the counterparty into the context)

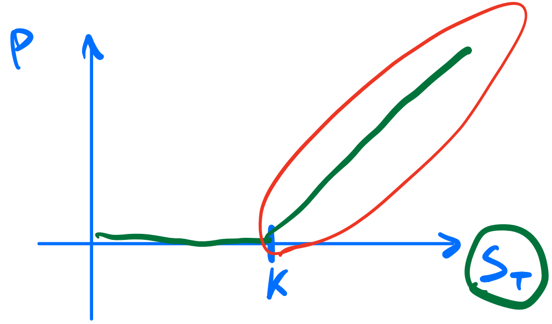
Derivatives Hedging

Lecture 6, Lecture 9

Derivatives Hedging

Imagine that we have a derivative — A [non-linear] function that depends on the [path of] price of some financial instrument[s]

- We want to train a policy $\pi(s)$ to hedge as good as possible the derivative
- **A common misconception**: as ~~good as possible refers to expected return~~



$\Delta \cdot \mu \sim \Delta \cdot \mu$

$\max(S_T - K, 0)$

Intermezzo: Risk-Neutral Pricing

The price of the derivative is the present value of the expected payoff under risk-neutral measure [see Fundamental Theorem of Asset Pricing]:

$$P_t = B_t \mathbb{E}^{\mathbb{Q}}[\frac{V_T}{B_T} \,|\, \mathcal{F}_t]$$

Under risk-neutral pricing the expected value of any stock portfolio (in particular, the hedging portfolio D_t) is just the risk-free rate, so your total portfolio has expected value of zero, **no matter what your RL agent learns**

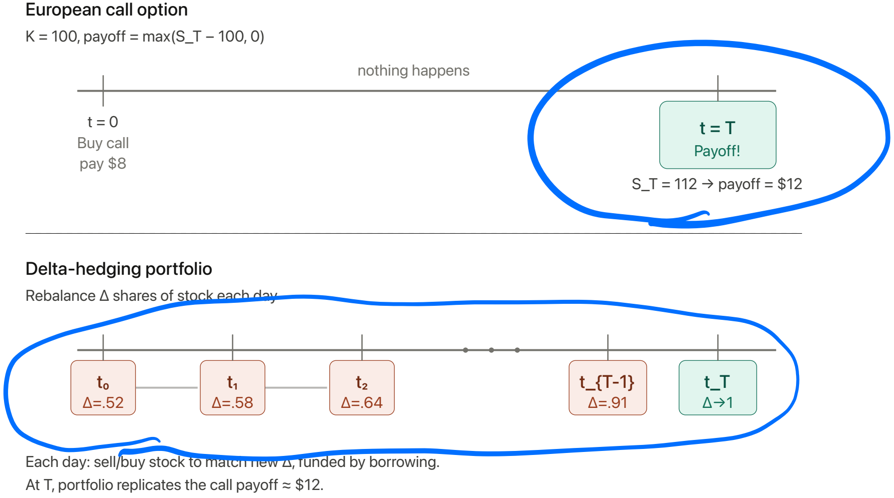
$$\mathbb{E}^{\mathbb{Q}}[\frac{V_T}{B_T} + \frac{D_T}{B_T} \,|\, \mathcal{F}_t] - \frac{P_t}{B_t} = 0$$

So, our actual RL task is w.r.t. the risk measure (e.g., standard deviation of such a portfolio)

Derivatives Hedging

So, our actual RL task is w.r.t. the risk measure (e.g., standard deviation of such a portfolio) $V_N(V_T - P(\pi(s))) \rightarrow \min_{\pi}$

Challenge — time mismatch and non-additive variance:



Source: Claude

Derivatives Hedging

Ways to approach this task:

- Model hedging weights as one path $(w_0, \ldots, w_T)$ and evaluate on the final metric only $(w_0, \ldots, w_T) \to \rho((w_0, \ldots, w_T)) \implies$ **supervised learning task** [see Buehler et al., 2018]
- **Potential function shaping** and Model-Based RL

Advantages of RL:

- Can work with simulated environments here, because under risk-neutral measure there is no need for risk premium simulation, which is **extremely hard**

Background

1. Dixon, Halperin, Bilokon — ML in Finance, Chapters 9, 10, 11 Book
2. Overview of RL in Finance Methods Paper
3. Mathematics for New Technologies in Finance Course

Thank you for your attention!